

AP Precalculus — Unit 2 Exam

Topics 2.7–2.14

Time: 45 minutes

Name: _____

Date: _____

Section I: Multiple Choice (8 Questions)

1. Let $f(x) = 3x + 2$ and $g(x) = x^2 - 1$. What is $(f \circ g)(x)$?
(A) $3x^2 + 1$
(B) $3(x^2 - 1) + 2$
(C) $3x^2 - 1$
(D) $x^2 + 5$
2. Which function has an inverse on its entire domain?
(A) $f(x) = x^2$
(B) $f(x) = |x|$
(C) $f(x) = x^3$
(D) $f(x) = \sin x$
3. Simplify: $\log_2(8x^4)$
(A) $3 + 4\log_2 x$
(B) $4 + 3\log_2 x$
(C) $\log_2 8 + \log_2 x^4$
(D) $8\log_2 x^4$
4. Condense: $\log x - \log y$
(A) $\log(x - y)$
(B) $\log\left(\frac{x}{y}\right)$
(C) $\log(xy)$
(D) $\frac{\log x}{\log y}$
5. Solve: $\ln(x - 2) = 1$
(A) $x = 1$
(B) $x = e + 2$
(C) $x = \ln 3$
(D) $x = e - 2$
6. Solve: $e^{2x} = 5$
(A) $\ln 5$
(B) $\frac{\ln 5}{2}$
(C) $2\ln 5$
(D) $e^{5/2}$
7. Solve: $\log(x^2 - 4) = 1$
(A) $x = \pm 5$
(B) $x = \pm\sqrt{14}$
(C) $x = \pm\sqrt{6}$
(D) $x = 5$
8. The graph of $f(x) = \log x$ is:
(A) Increasing and concave up
(B) Increasing and concave down
(C) Decreasing and concave up
(D) Decreasing and concave down

Section II: Free Response (3 Questions)

Calculator is allowed. Show all work.

Question 1

A function f is defined by the table below.

x	-1	0	1	2
$f(x)$	12	6	3	1.5

The function g is defined by $g(x) = \log_2(x)$.

- (A) Define $h(x) = (g \circ f)(x)$. Find $h(1)$ or indicate that it is not defined.
- (B) Determine whether f is linear, quadratic, exponential, or logarithmic. Justify your answer using the table.
- (C) Find the inverse of g and state its domain.

Question 2

The number of users of an app is modeled by

$$N(t) = 50(1.12)^t$$

where t is measured in months and $N(t)$ is in thousands.

(A) Interpret the meaning of $N(0)$ in context.

(B) Find the average rate of change of $N(t)$ from $t = 0$ to $t = 6$.

(C) Determine whether the growth is linear or exponential and explain why.

Question 3

Let $f(x) = \log_3(x - 1)$.

(A) State the domain of f .

(B) Solve the inequality $\log_3(x - 1) \geq 2$.

(C) Explain how the graph of f is related to the graph of $y = \log_3 x$.